Capstone controlled experiments

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

Before you start

What must stay fixed when comparing learning rates?

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Trace matched two-update learning-rate experiments. Try the recall before opening notes.

Your learning target

I can trace matched two-update learning-rate experiments.

I can select a candidate using a prespecified validation rule.

I can explain the limits of a one-factor comparison.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Capstone hypothesis and baseline. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Efficiency and resource measurement. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Transfer learning. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Trace matched two-update learning-rate experiments

Select a candidate using a prespecified validation rule

Explain the limits of a one-factor comparison

Retrieve what you already know

> What must stay fixed when comparing learning rates?

My starting explanation

This week's end goal

Explain capstone controlled experiments with a traceable toy-network case and a bounded conclusion.

Notice the evidence

> Investigate the supplied Capstone controlled experiments case. Separate observations, given facts and assumptions before choosing an explanation.

Continue with **training** (1,2) and (2,4), validation (3,6) and (4,8), and **prediction** max(0, a*x). Compare learning rates 0.1 and 0.2. Each run starts at a = 1, has no bias and takes two full-batch updates.

Keep the loss, data order and arithmetic fixed. This toy model has no randomness; repeating the exact run checks replay, not variation across seeds.

Record the comparison as an explicit revision to the one-candidate Week 33 plan before viewing new results. Validation may guide selection; sealed final data must not.

Select lower validation MSE, with a tie going to learning rate 0.1.

> **List the given facts and one fact that is not supplied.**

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Capstone controlled experiments. Connect each labelled input, deciding step and result.

For positive weights and these positive **input**s, the ReLU is active. **Training** MSE is ((a-2)^2 + (2*a-4)^2)/2 = 2.5*(a-2)^2. Its gradient is 5*(a-2). Update a to a - rate*5*(a-2). At rate 0.1, weights are 1.5 then 1.75.

At rate 0.2, weights are 2 then 2. The formula assumes the active region; do not extend it blindly to nonpositive weights.

Run	Final weight	Training MSE	Validation MSE
Baseline	1	2.5	12.5
Rate 0.1, two updates	1.75	0.15625	0.78125
Rate 0.2, two updates	2	0	0

Keep the evidence roles separate

PLAN OR INPUT	PROCEDURE	PERMITTED CONCLUSION
Same initial a = 1	Rate 0.1, two updates	Validation MSE 0.78125
Same initial a = 1	Rate 0.2, two updates	Validation MSE 0
Same validation rows	Choose lower MSE	Select rate 0.2

The selected run fits this exact constructed relationship. Its zero error does not establish universal reliability.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

Compare the cases

> Compare the supplied cases for Capstone controlled experiments. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

Compute both updates for rates 0.1 and 0.2 using gradient 5*(a-2).

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Change one thing in the pictured case

Change one thing

Change: Change B's **training** set as well as the feature connection.

Predicted result

Prediction: The experiment no longer isolates the connection change.

Why the result follows

Reason: A **training**-data difference supplies an alternative explanation for the observed result.

Words that unlock the idea

> Use the vocabulary of Capstone controlled experiments to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Controlled comparison	A comparison holding specified conditions constant while changing an intended factor
Full batch	An update whose gradient uses all training rows
Selection rule	A rule for choosing a candidate using development evidence
Confounder	A differing condition that prevents isolating the intended explanation

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Capstone controlled experiments, predict a step, then compare with its explanation.

Worked explanation · Capstone controlled experiments

At a = 1.75, validation **prediction**s are 5.25 and 7. Errors -0.75 and -1 square to 0.5625 and 1, giving MSE 0.78125. At a = 2 the **prediction**s equal 6 and 8, so MSE is zero. Rate 0.2 wins the written validation rule.

This selection uses two validation cases, not four independent cases because two runs used them.

Worked explanation · Capstone controlled experiments

If one run used extra epochs or different **training** rows, the comparison would mix factors. Either rerun a matched experiment or report the differences and avoid claiming an isolated learning-rate effect.

A zero validation score is also compatible with a very simple task and says nothing about a new distribution.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Capstone controlled experiments

One controlled network change. The capstone compares two fixed networks on the same six labelled toy patterns. Only one feature connection is added.

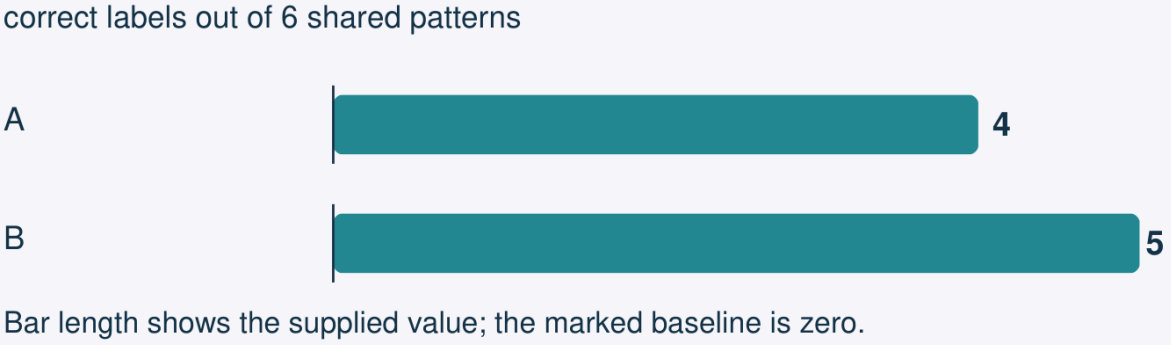
The facts we will use

Supplied facts

A scores [1,1,0,1,0,1]; B scores [1,1,1,1,0,1]. Settings, labels and scoring remain fixed. Binary entries mean 1=correct and 0=incorrect.

AI ACADEMY / AI-5/34

Controlled toy-pattern comparison



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/34

Capstone controlled experiments

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Identify the intervention

B adds the centre-feature connection; all other declared settings stay fixed.



2. Count outcomes

A is 4/6 correct; B is 5/6.



3. Inspect paired cases

Only case three changes correctness; case five fails for both.

CHANGE ONE THING

Change B's training set as well as the feature connection.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Identify the intervention	B adds the centre-feature connection; all other declared settings stay fixed.
2. Count outcomes	A is 4/6 correct; B is 5/6.
3. Inspect paired cases	Only case three changes correctness; case five fails for both.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Controlled experiments vary the specified component while retaining comparable cases. Pairing the outcomes by pattern shows where the score difference comes from instead of hiding it in an overall average.

What this example does not establish

Keep the stated limit

These descriptive toy results do not establish significance, stable superiority or the full learned **mechanism**.

Keep the stated limit

Read the labelled picture: Compute the two correct counts and identify the changed case. Why would changing B's **training** data weaken the conclusion about its added connection?

Keep the stated limit

Change one thing: Change B's **training** set as well as the feature connection. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Capstone controlled experiments claim and explain a repair using the supplied evidence.

Supported example: rate 0.3 starts at a = 1. The first gradient is -5, giving a = 2.5. The next gradient is 2.5, giving a = 1.75. This two-update run finishes at the same weight as rate 0.1 although its path differs.

Guided: calculate the second update and its validation MSE. Explain why equal final scores do not mean identical trajectories.

Locate the earliest unsupported step

Recompute validation MSE for the final weights and apply the written **selection rule**. How many distinct validation rows were used?

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

Practise Capstone controlled experiments with support, then explain the deciding step in your own words.

List six settings held fixed and the single intended changed factor.

Guided: calculate the second update and its validation MSE. Explain why equal final scores do not mean identical trajectories.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace both runs

Use workbook W2 for the connected example “Capstone controlled experiments”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Capstone controlled experiments. Proposed explanation: “More runs mean more validation examples”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Compute the two correct counts and identify the changed case. Why would changing B’s **training** data weaken the conclusion about its added connection?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Capstone controlled experiments. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

Compute both updates for rates 0.1 and 0.2 using gradient 5*(a-2).

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Capstone controlled experiments. Explain why each step is justified.

Explain → apply → challenge

List six settings held fixed and the single intended changed factor.

> Recompute validation MSE for the final weights and apply the written **selection rule**. How many distinct validation rows were used?

Run A uses rate 0.1 for two updates; run B uses rate 0.2 for six updates and different **training** rows. Can the difference be attributed to rate alone? Propose a repair.

> **Explain which evidence supports your conclusion and which claim remains unproved.**

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Image-pattern Network Investigation. Keep an annotated decision and the evidence supporting it.

For Image-pattern Network Investigation, use this week's independent task as a rehearsal for the final **evidence** pack: For **training** (1,3),(2,6) and validation (3,9),(4,12), start a = 1.

Compare rates 0.1 and 0.2 for two updates. Gradient is 5*(a-3). Compute final weights and validation MSE, then select lower MSE; ties go to 0.1.

Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Save run IDs, settings, each checkpoint, **prediction**s and metric calculations. Report the losing run as well as the winner. Freeze the selected rate, architecture and update count before the final evaluation.

Any change after opening final data requires a new development record and fresh reserved **evidence**.

> **Exit: one run receives six updates and the other two. What can the score comparison establish, and what must be rerun to isolate the learning rate?**

Run A uses rate 0.1 for two updates; run B uses rate 0.2 for six updates and different **training** rows. Can the difference be attributed to rate alone? Propose a repair.

> **My limited claim and a useful next check**

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Capstone controlled experiments case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Keep the evidence roles separate

PLAN OR INPUT	PROCEDURE	PERMITTED CONCLUSION
Same initial a = 1	Rate 0.1, two updates	Validation MSE 0.78125
Same initial a = 1	Rate 0.2, two updates	Validation MSE 0
Same validation rows	Choose lower MSE	Select rate 0.2

The selected run fits this exact constructed relationship. Its zero error does not establish universal reliability.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Run a new comparison independently

For **training** (1,3),(2,6) and validation (3,9),(4,12), start a = 1. Compare rates 0.1 and 0.2 for two updates. Gradient is 5*(a-3). Compute final weights and validation MSE, then select lower MSE; ties go to 0.1.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

For **training** (1,3),(2,6) and validation (3,9),(4,12), start a = 1. Compare rates 0.1 and 0.2 for two updates. Gradient is 5*(a-3). Compute final weights and validation MSE, then select lower MSE; ties go to 0.1.

> **My independent procedure, result and limitation**

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Capstone controlled experiments, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: One controlled network change. The capstone compares two fixed networks on the same six labelled toy patterns. Only one feature connection is added.

By the end, I can	How I will show it
Trace matched two-update learning-rate experiments	Explain the deciding step, show a trace or test, and state one limit.
Select a candidate using a prespecified validation rule	Explain the deciding step, show a trace or test, and state one limit.
Explain the limits of a one-factor comparison	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Capstone controlled experiments without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Trace matched two-update learning-rate experiments

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

One controlled network change. The capstone compares two fixed networks on the same six labelled toy patterns. Only one feature connection is added.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Only case three changes correctness; case five fails for both.

1. Inspect paired cases

2. Identify the intervention

3. Count outcomes

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Inspect paired cases Only case three changes correctness; case five fails for both.

Controlled experiments vary the specified component while retaining comparable cases. Pairing the outcomes by pattern shows where the score difference comes from instead of hiding it in an overall average.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change B's **training** set as well as the feature connection. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. The experiment no longer isolates the connection change.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The experiment no longer isolates the connection change. A **training**-data difference supplies an alternative explanation for the observed result.

Repair a missing building block

Review Capstone hypothesis and baseline (ai-5/33). Goal: Write a testable capstone hypothesis and a credible baseline. Short practice: Name the intended change, the baseline and the measured outcome.

Why must both versions avoid **training** on the eight reserved labels?. Return to this week and explain what you changed.

Review Efficiency and resource measurement (ai-5/31). Goal: Calculate throughput and amortised time per example from repeated timings. Short practice: Why is 10ms/case not necessarily a single-request latency?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Transfer learning (ai-5/28). Goal: Trace a reused feature extractor and a newly fitted **output** head. Short practice: Which part was trained for the new task?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Keep your completed work

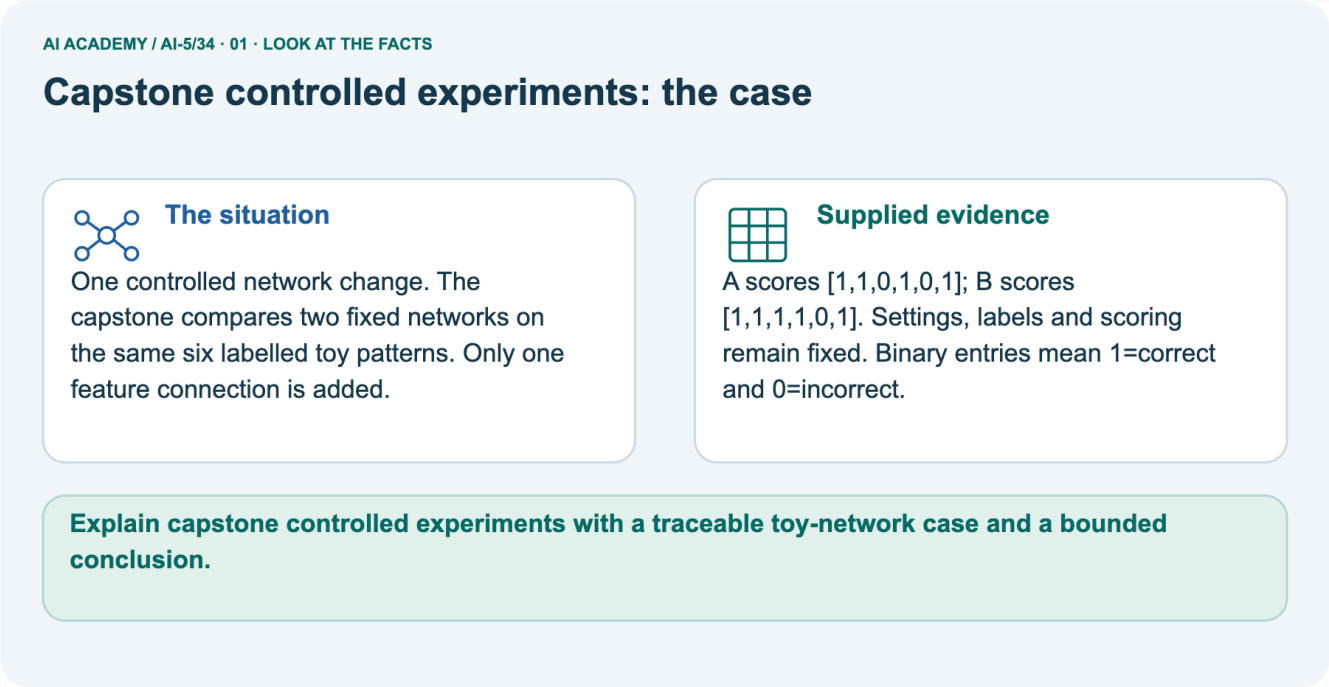
Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

Visual concept journey

> Explain each Capstone controlled experiments graphic using its labels, deciding evidence and relevant limitation.

01. Capstone controlled experiments: the case



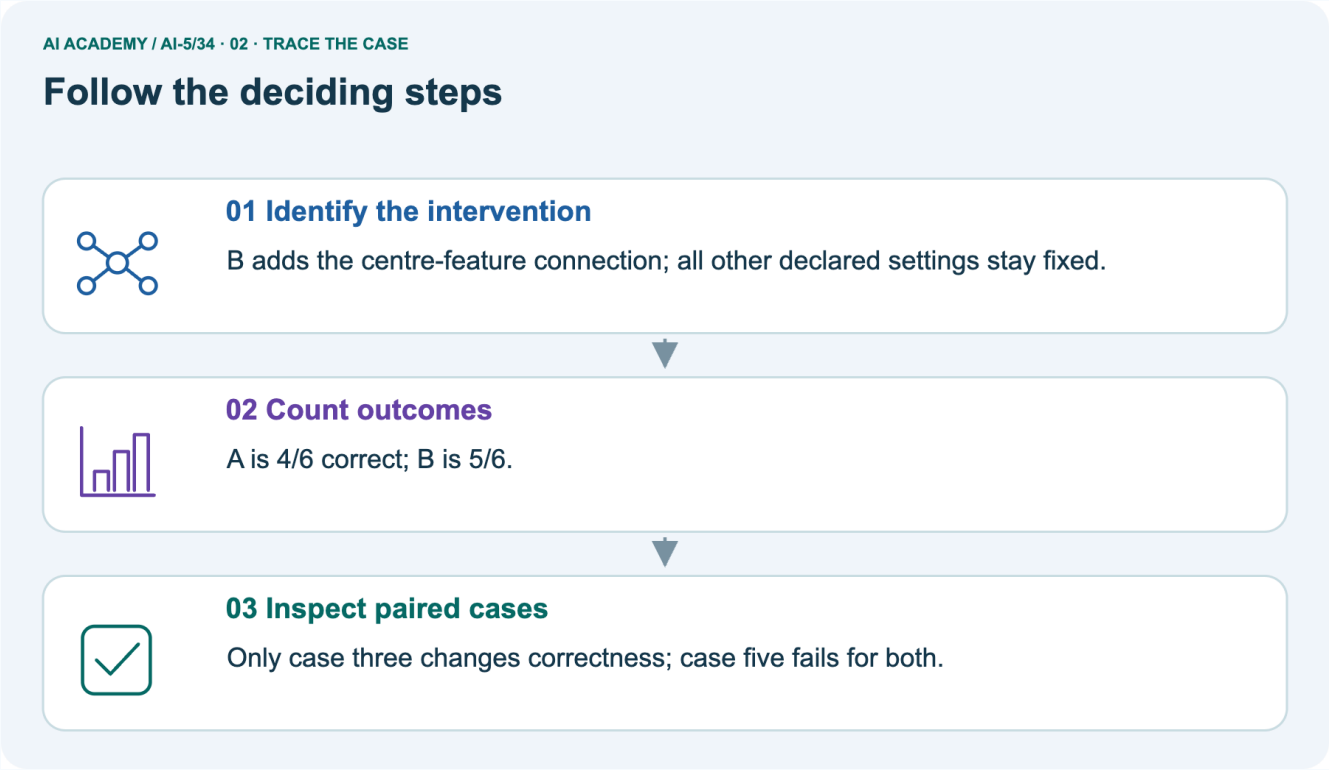
Follow the labelled picture with the matching explanation. Open the vector diagram

One controlled network change. The capstone compares two fixed networks on the same six labelled toy patterns. Only one feature connection is added. Supplied facts: A scores [1,1,0,1,0,1]; B scores [1,1,1,1,0,1].

Settings, labels and scoring remain fixed. Binary entries mean 1=correct and 0=incorrect.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

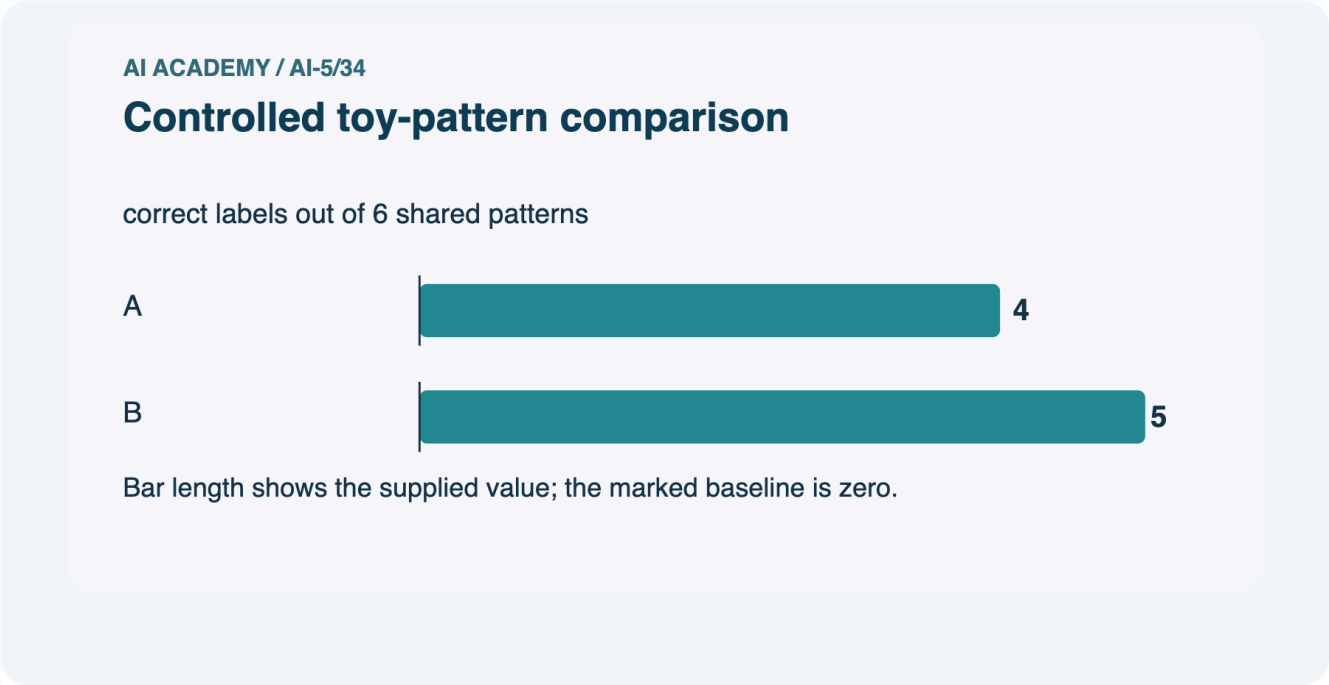


Follow the labelled picture with the matching explanation. Open the vector diagram

Identify the intervention: B adds the centre-feature connection; all other declared settings stay fixed. → Count outcomes: A is 4/6 correct; B is 5/6. → Inspect paired cases: Only case three changes correctness; case five fails for both.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Controlled toy-pattern comparison

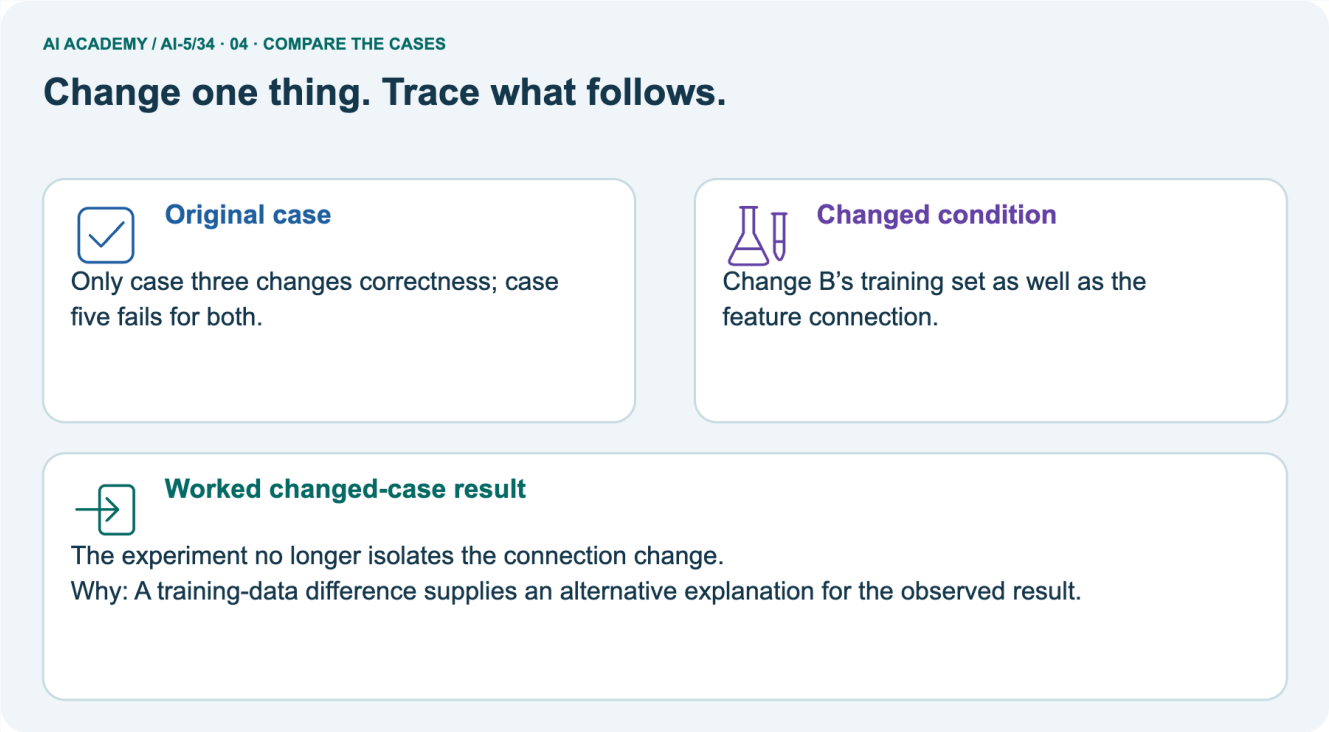


Follow the labelled picture with the matching explanation. Open the vector diagram

Controlled toy-pattern comparison. Units: correct labels out of 6 shared patterns. A: 4. B: 5.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



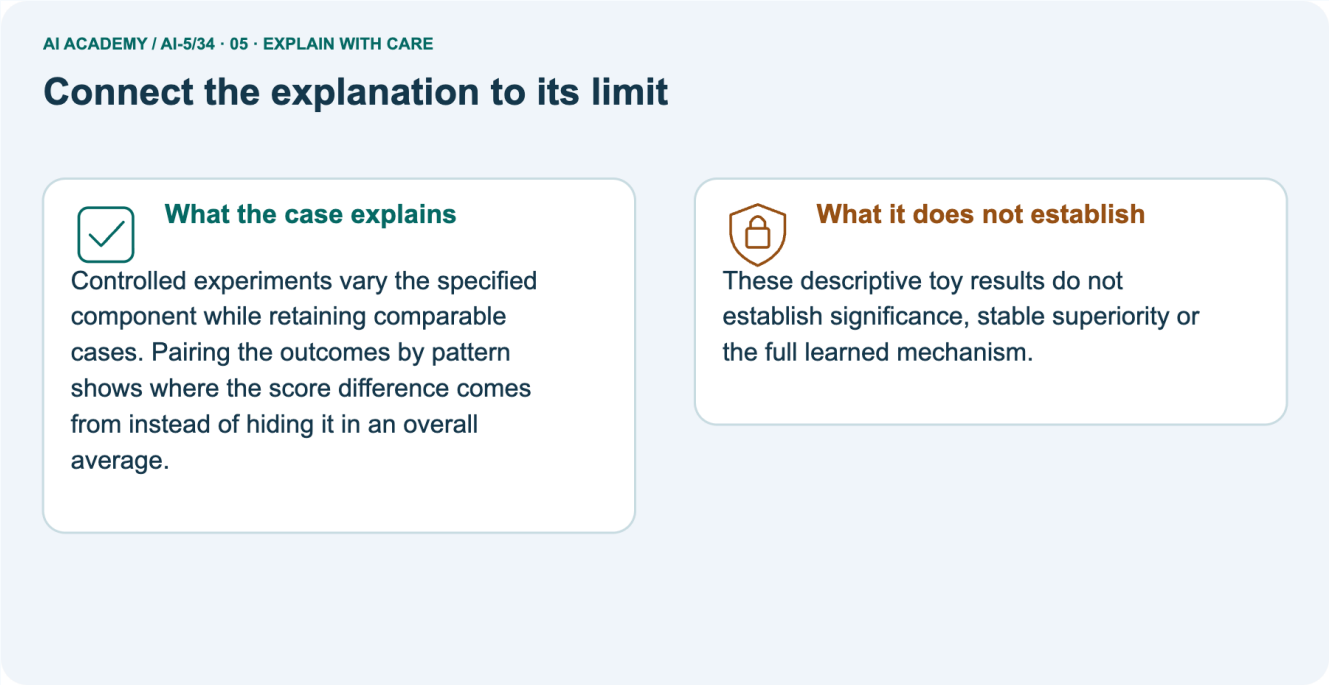
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Only case three changes correctness; case five fails for both. Changed condition: Change B's **training** set as well as the feature connection.

Expected result: The experiment no longer isolates the connection change. Reason: A **training**-data difference supplies an alternative explanation for the observed result.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



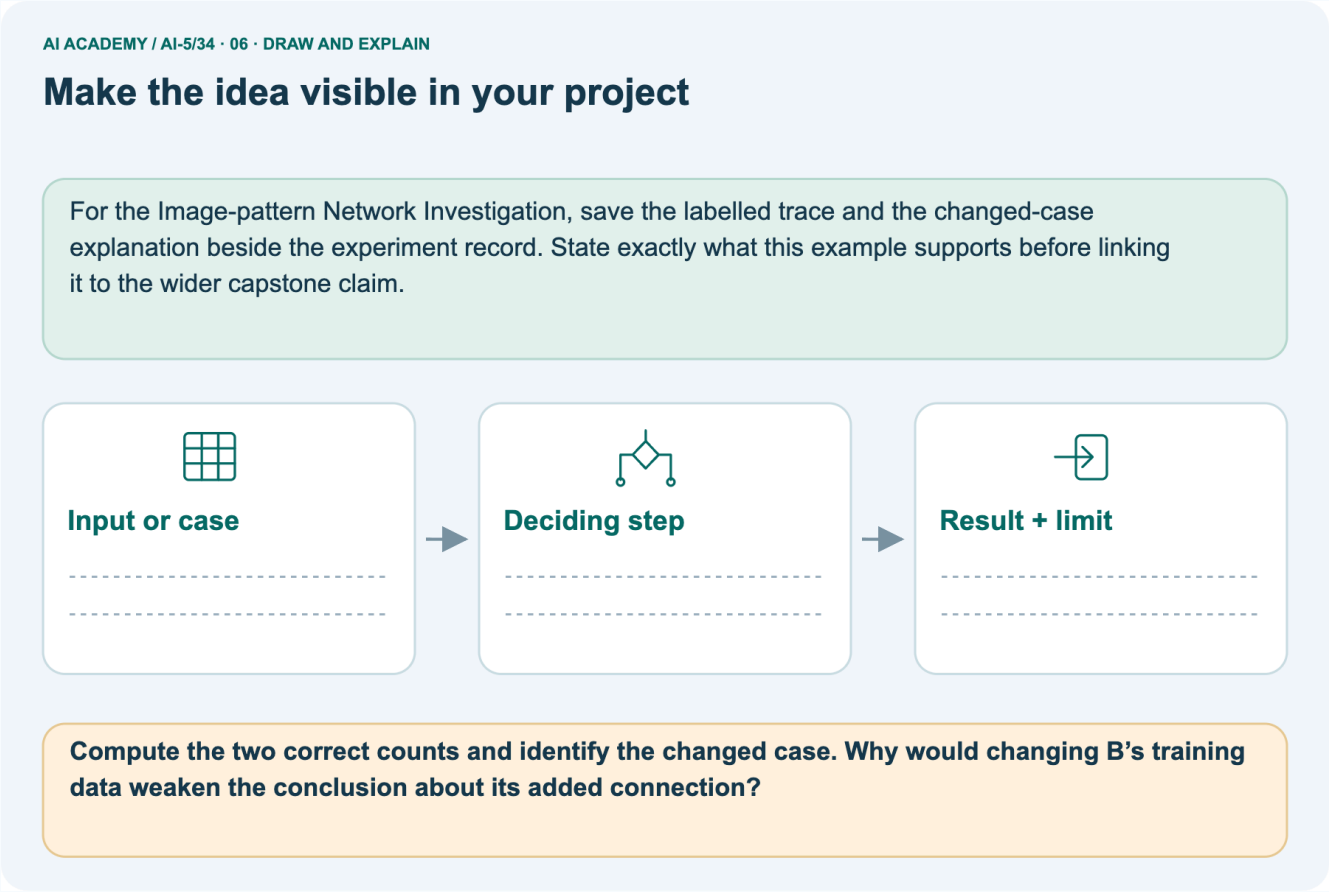
Follow the labelled picture with the matching explanation. Open the vector diagram

Controlled experiments vary the specified component while retaining comparable cases. Pairing the outcomes by pattern shows where the score difference comes from instead of hiding it in an overall average.

Limit: These descriptive toy results do not establish significance, stable superiority or the full learned **mechanism**.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

Draw your **input**, deciding step, result and limit. Supported question: Compute the two correct counts and identify the changed case. Why would changing B's **training** data weaken the conclusion about its added connection?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.